

Green University of Bangladesh

Department of Computer Science and Engineering (CSE)
Semester: (Fall, Year: 2023), B.Sc. in CSE (Day)

Tower Bridge - An Architectural Marvel

Course Title: Engineering Drawing Lab
Course Code: CSE 208
Section: 221-D2

Students Details

Name	ID
Md.Fahad Nakib	221902149
Md.Nafijul Islam	221902156

Submission Date: 14-12-23
Course Teacher's Name: Md Naimul Pathan.

[For teachers use only: Don't write anything inside this box]

Lab Project Status	
Marks:	**Signature:**
Comments:	**Date:**

Contents

1 Introduction **3**

1.1 Overview . . . 3

1.2 Motivation . . . 3

1.3 Problem Definition . . . 3

1.3.1 Problem Statement . . . 3

1.3.2 Tools and Technology . . . 4

1.3.3 Complex Engineering Problem . . . 4

1.4 Design Goals/Objectives . . . 4

1.5 Conclusion . . . 5

2 Design/Development/Implementation of the Project **6**

2.1 Introduction . . . 6

2.2 Project Details . . . 6

2.2.1 Data Collection: . . . 6

2.2.2 Software Selection . . . 6

2.2.3 Modeling . . . 6

2.2.4 Texturing and Materials . . . 7

2.2.5 Lighting Simulation . . . 7

2.3 Implementation . . . 7

2.3.1 Insert Image . . . 7

3 Performance Evaluation **12**

3.1 Results Analysis . . . 12

3.1.1 Final Project . . . 12

3.1.2 Corner View . . . 13

3.1.3 Corner View . . . 13

4 Conclusion **14**

4.1 Discussion 14
4.2 Scope of Future Work 14

Chapter 1

Introduction

1.1 Overview

This project aims to develop a comprehensive and visually stunning three-dimensional (3D) model of the iconic Tower Bridge in London. The primary objective is to create an immersive digital representation that accurately reflects the historical and architectural significance of Tower Bridge. The 3D model will serve as an educational and promotional tool, offering a virtual experience for users to explore and learn about one of London's most renowned landmarks.

1.2 Motivation

The main motivations behind creating a 3D model of Tower Bridge extend beyond the realm of technology and design. They encompass cultural preservation, education, tourism, and community engagement, contributing to a holistic appreciation of this iconic landmark.

1.3 Problem Definition

1.3.1 Problem Statement

Develop a highly detailed and accurate 3D model of Tower Bridge. Incorporate historical and architectural details to showcase the evolution of the bridge's design. Create an interactive virtual tour for users to explore the bridge from various angles and perspectives. Enhance public awareness and appreciation of Tower Bridge's historical and cultural importance.

1.3.2 Tools and Technology

Creating a 3D model of the Tower Bridge involves using various tools and technologies. The process typically includes a combination of 3D modeling software, geographic information systems (GIS), and sometimes specialized hardware. Here are some tools and technologies

CAD Software:

AutoCAD: Often used in architecture and engineering, AutoCAD can be used for precise 2D and 3D drafting.

3D Modeling Software:

Autodesk Maya: A powerful 3D modeling and animation software. Blender: An open-source 3D creation suite that supports modeling, sculpting, animation, and more. SketchUp: A user-friendly 3D modeling tool, particularly good for architectural modeling.

1.3.3 Complex Engineering Problem

Table 1.1: Summary of the attributes touched by the mentioned projects

Name of the P Attributess	**Explain how to address**
P1: Research Phase:	Gather historical data, architectural blueprints, and high-resolution images of London Bridge.
P2: Modeling Phase	Use industry-standard 3D modeling software to create an accurate representation of the bridge, ensuring attention to detail.
P3: Texturing and Rendering:	Apply realistic textures and lighting to enhance the visual appeal of the 3D model.

1.4 Design Goals/Objectives

Develop a highly detailed and accurate 3D model of Tower Bridge.
Incorporate historical and architectural details to showcase the evolution of the bridge's design.
Create an interactive virtual tour for users to explore the bridge from various angles and perspectives.
Enhance public awareness and appreciation of Tower Bridge's historical and cultural importance.

1.5 Conclusion

The 3D drawing of London Bridge project represents an exciting opportunity to blend technology, history, and art to create an engaging and informative experience. This initiative aligns with the growing interest in virtual tourism and educational content, contributing to a deeper understanding and appreciation of London's rich cultural heritage

Chapter 2

Design/Development/Implementation of the Project

2.1 Introduction

This project focuses on the 3D design of Tower Bridge, employing advanced computer-aided design (CAD) and 3D modeling techniques to create a detailed digital representation of this iconic structure. The objective is to leverage modern technology to enhance our understanding of the bridge's architectural complexities and explore potential design modifications.

2.2 Project Details

2.2.1 Data Collection:

Gathered historical blueprints, photographs, and architectural documents to ensure accuracy in the digital representation. Utilized laser scanning technology to capture precise measurements and dimensions of the physical structure.

2.2.2 Software Selection

Employed industry-standard 3D modeling software such as AutoCAD, Rhino, or SketchUp for creating the digital model. Utilized rendering software for realistic visualization.

2.2.3 Modeling

Modeled each component of Tower Bridge in 3D, considering architectural details, proportions, and structural elements. Ensured accuracy in scale and dimensions to replicate the real-world structure.

2.2.4 Texturing and Materials

Applied realistic textures and materials to mimic the bridge's appearance, considering the weathered look of historical structures. Emphasized intricate details such as stonework, metal surfaces, and ornate features.

2.2.5 Lighting Simulation

Simulated different lighting conditions to understand how the bridge would appear at various times of the day and under different weather conditions. Considered the interplay of natural and artificial lighting on the bridge's aesthetics.

2.3 Implementation

2.3.1 Insert Image

First of all, we have to insert blue print of Tower Bridge in our 3D drawing platform Auto Cad.

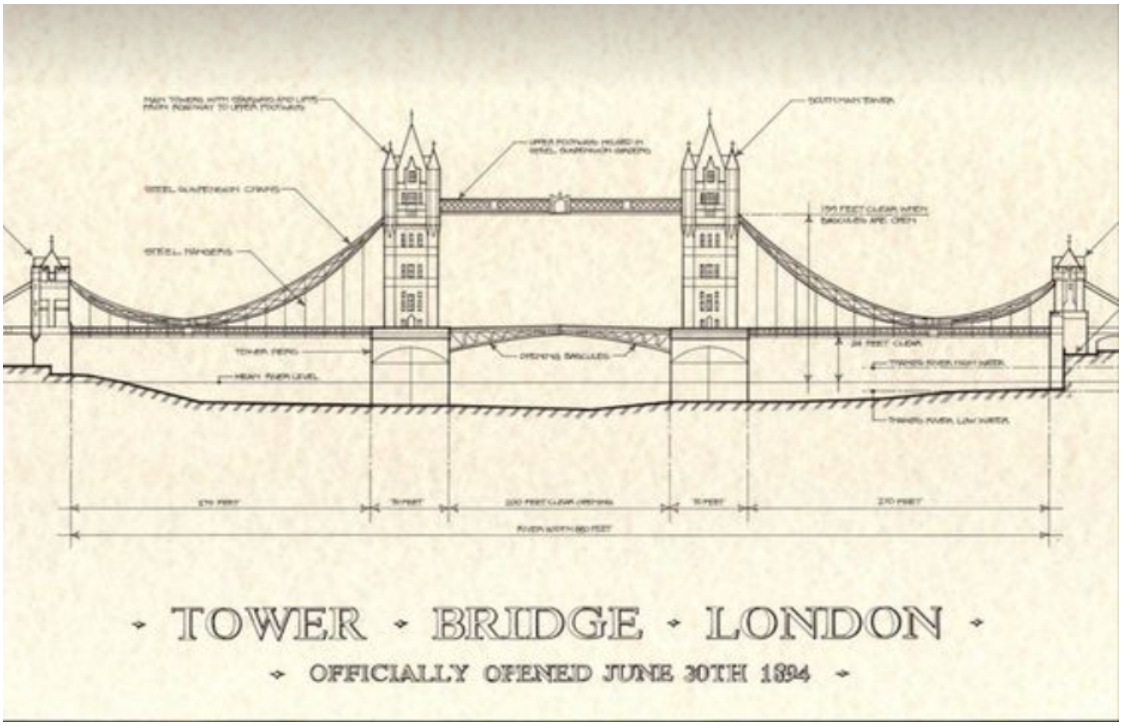


Figure 2.1: Blue print

The workflow

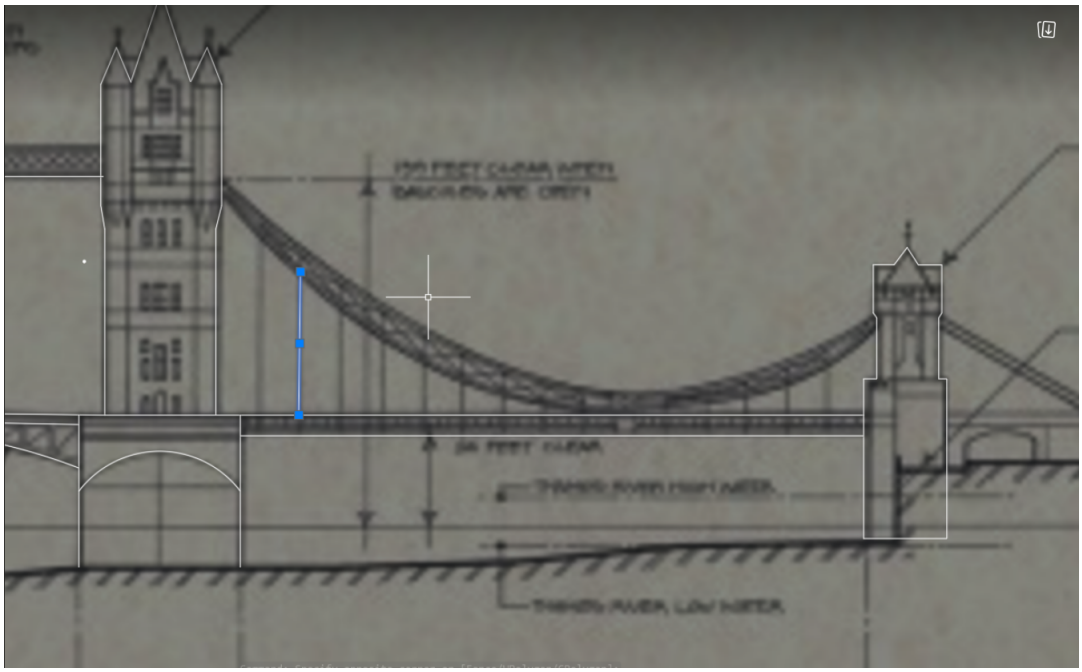


Figure 2.2: Taking measurement from image.

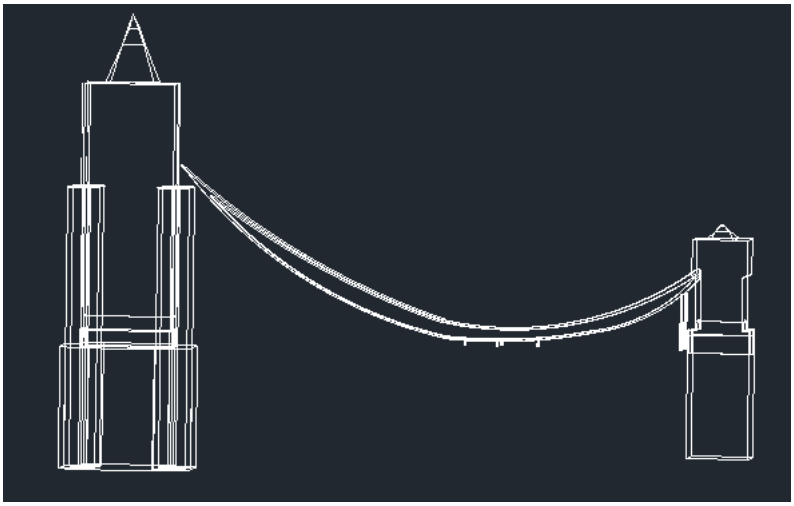


Figure 2.3: Taking measurement from image.

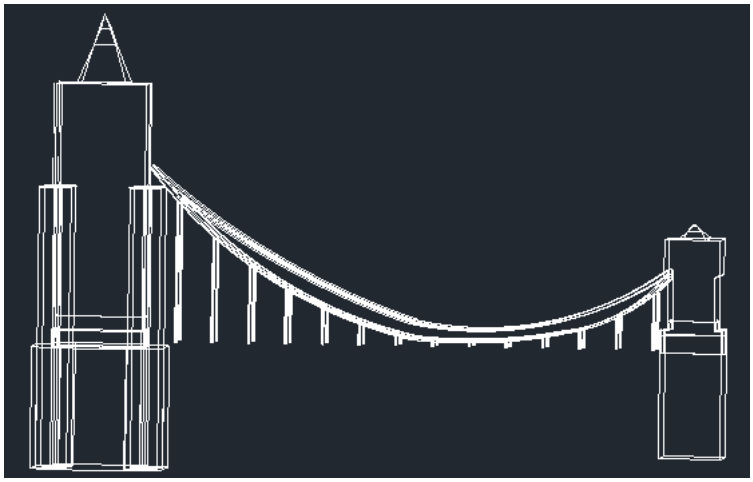


Figure 2.4: Taking measurement from image.

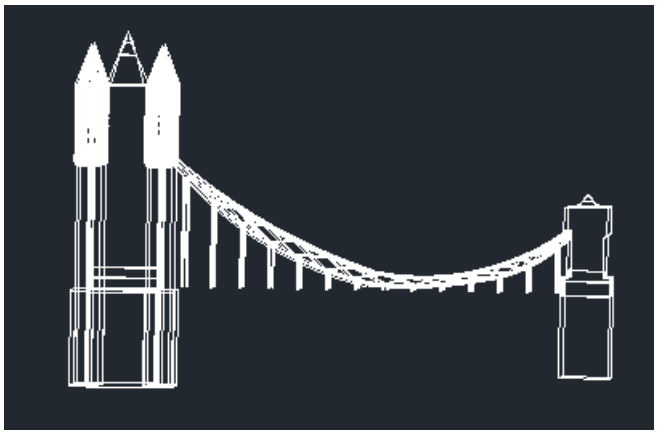


Figure 2.5: Taking measurement from image.

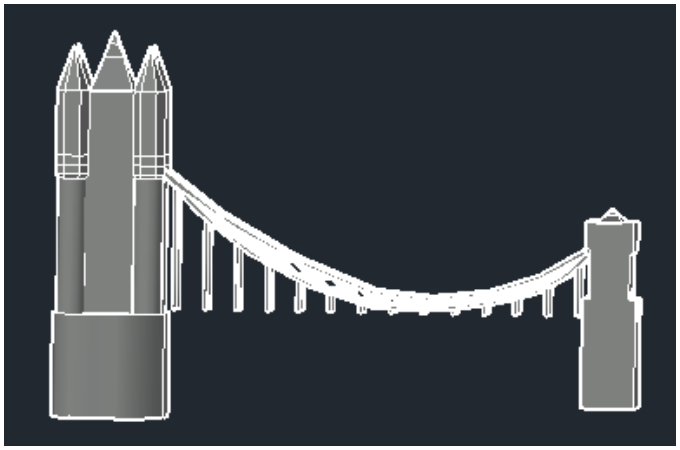


Figure 2.6: Creatig 3D view.

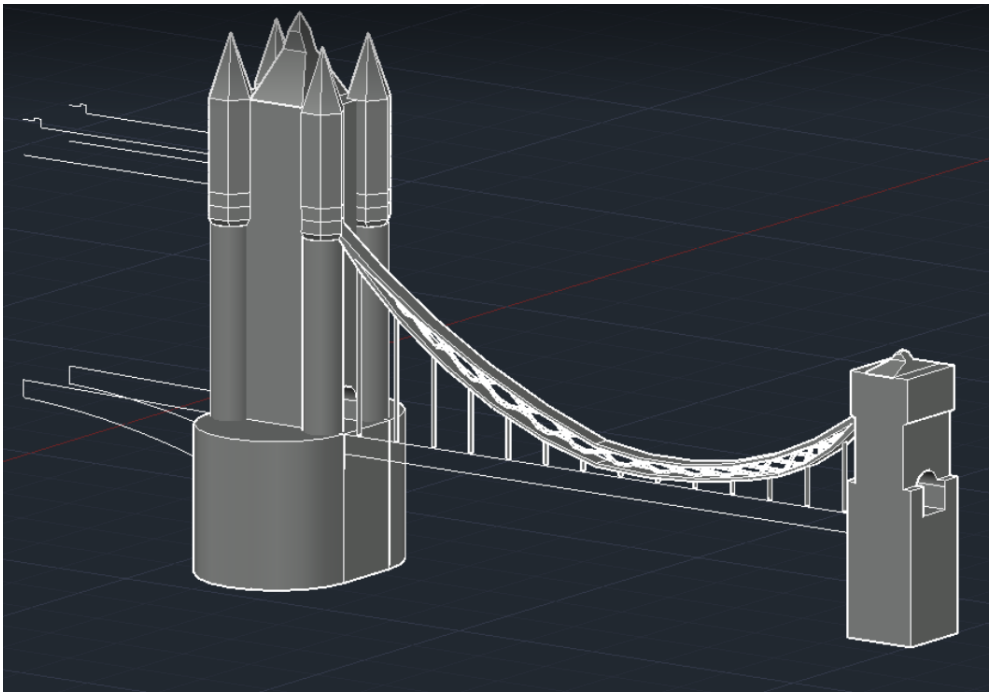


Figure 2.7: Creating Bridge .

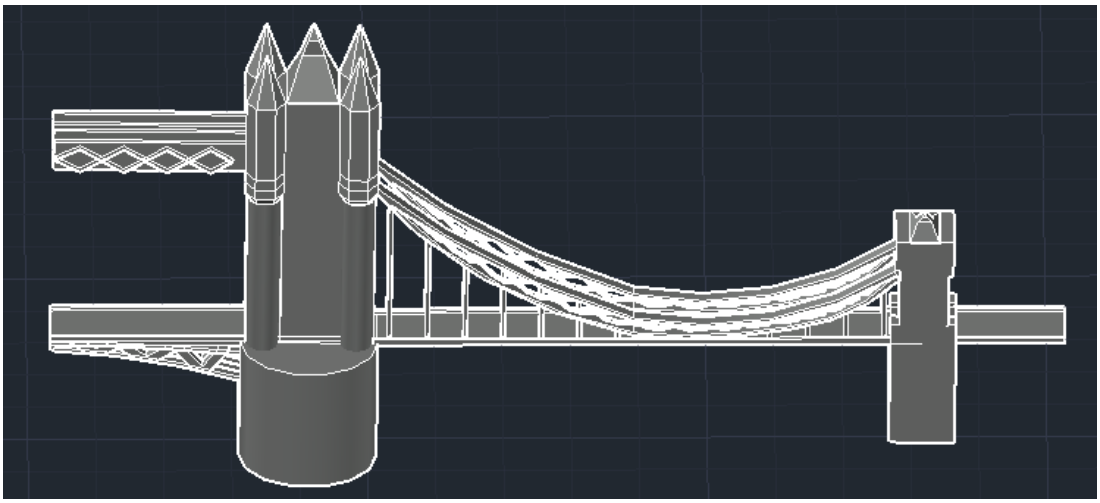


Figure 2.8: Using Extrude command create road.

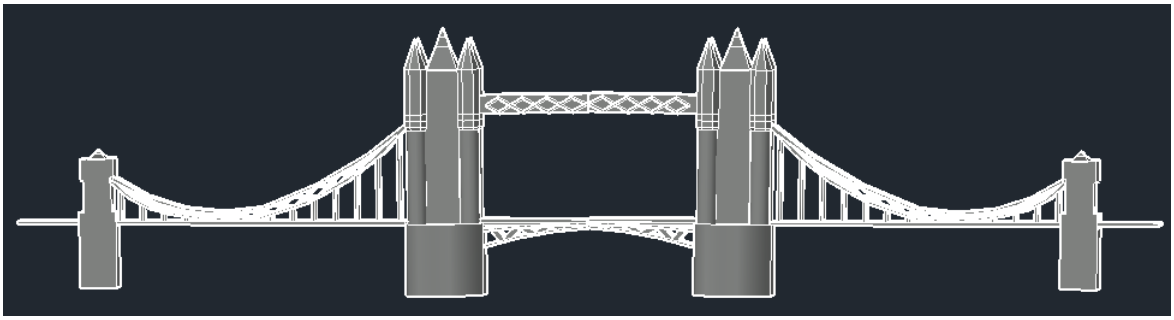


Figure 2.9: Using 3D mirror command create full bridge.

Chapter 3

Performance Evaluation

3.1 Results Analysis

3.1.1 Final Project

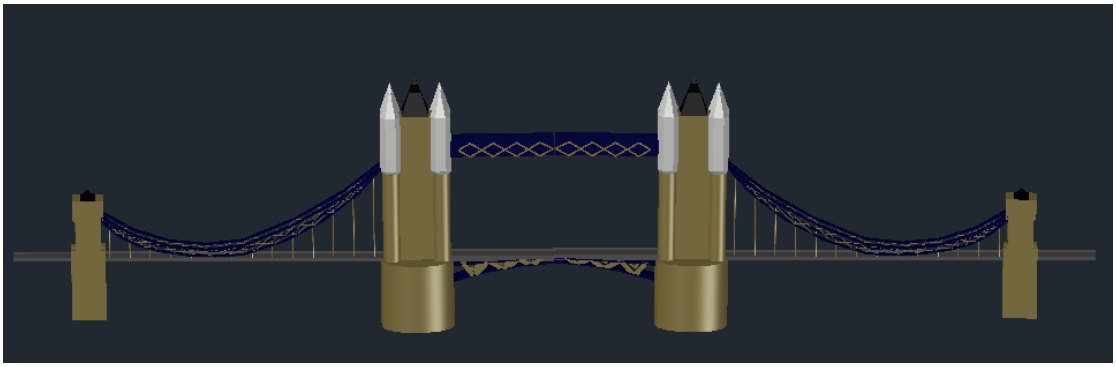


Figure 3.1: Final project

3.1.2 Corner View

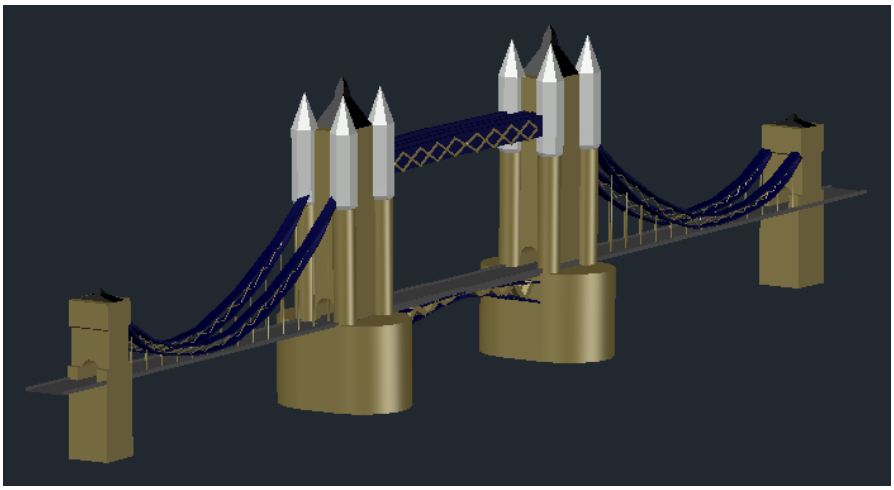


Figure 3.2: Corner View

3.1.3 Corner View

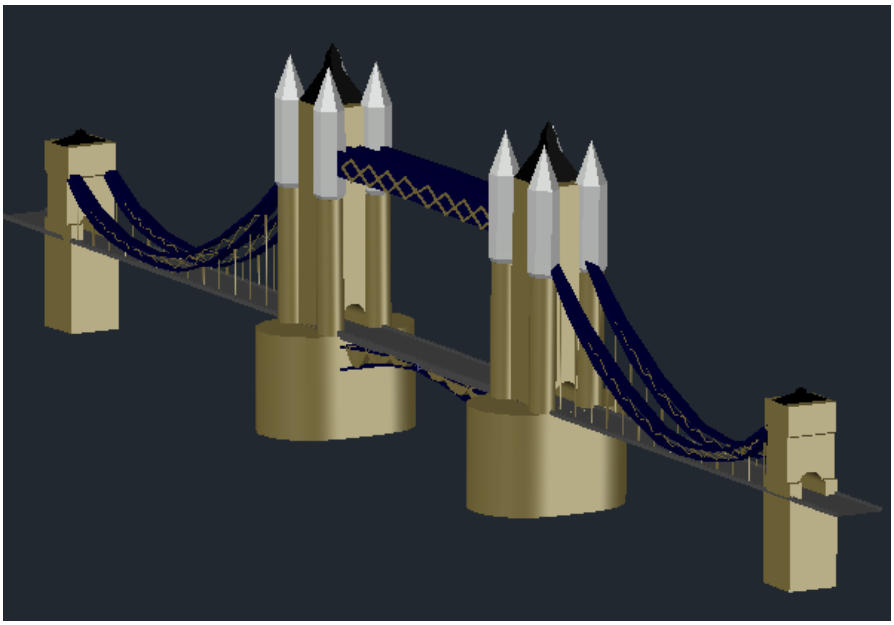


Figure 3.3: Corner View

Chapter 4

Conclusion

4.1 Discussion

The 3D design of Tower Bridge offers a novel approach to understanding and appreciating this architectural masterpiece. By combining historical research with cutting-edge technology, this project contributes to the preservation and dissemination of cultural heritage while providing valuable insights for future design endeavors.

4.2 Scope of Future Work

Integration with Augmented Reality (AR) and Virtual Reality (VR):

Explore the integration of the 3D model into AR and VR platforms for an even more immersive experience.

Collaborative Design:

Enable collaborative design processes, allowing multiple designers or stakeholders to contribute to the 3D model.